

ML-Based Loan Approval Prediction System

ACADEMIC PROJECT

MACHINE LEARNING

FINANCIAL TECHNOLOGY

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The Challenge: Manual Loan Processing

Delays

Manual reviews take days or weeks, frustrating applicants and slowing portfolio growth.

Inconsistency

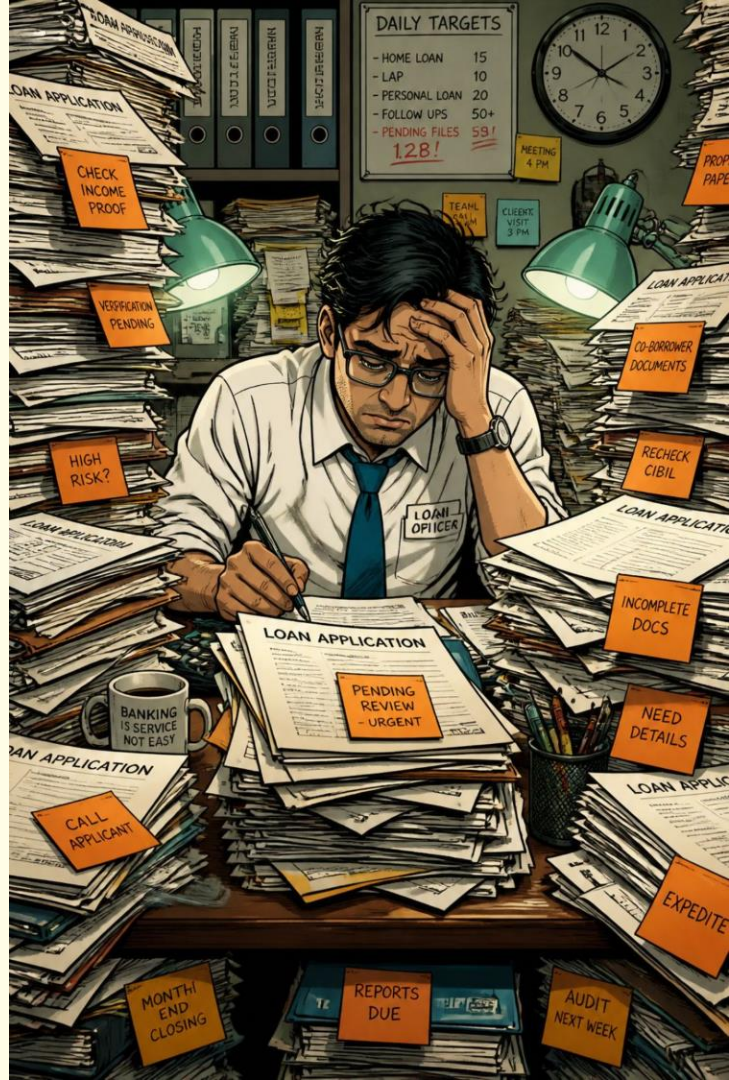
Human judgment varies by officer, branch, and time – producing unequal outcomes.

Errors

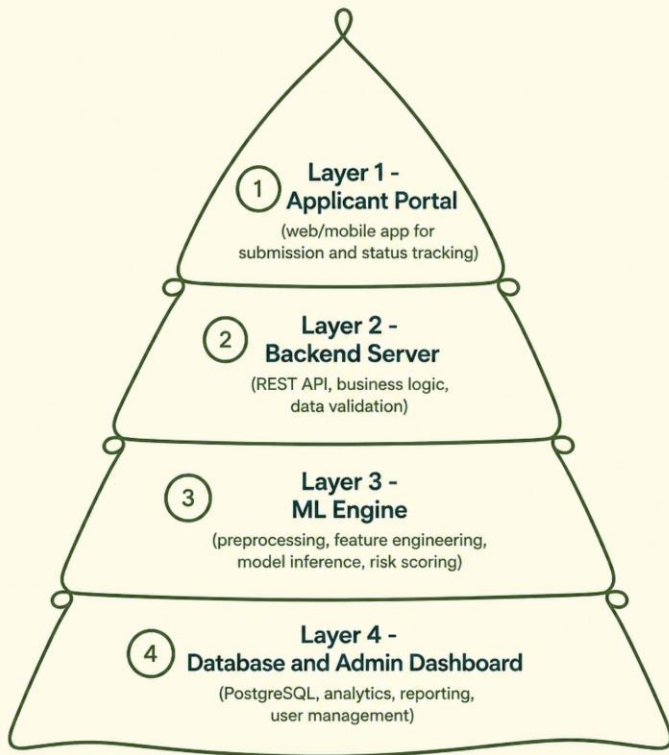
Data-entry mistakes and overlooked signals increase default risk and compliance exposure.

Poor Scalability

Traditional methods cannot handle high-volume application surges without major staffing costs.



System Architecture & Workflow



End-to-End Data Flow

The system operates as a closed-loop pipeline from submission to decision:

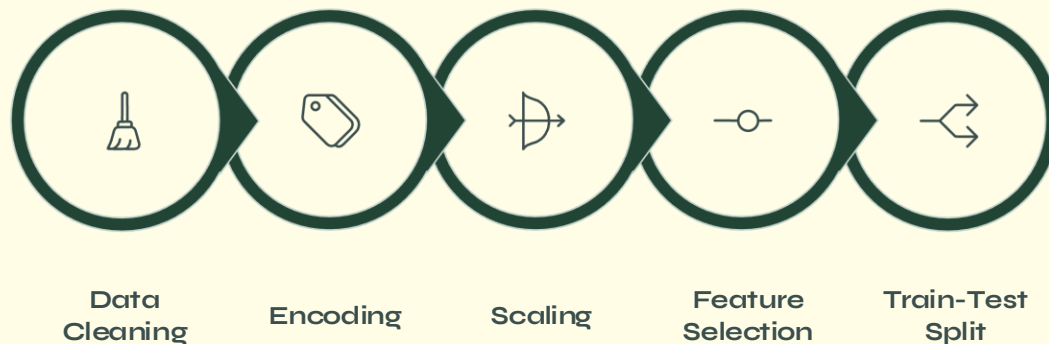
- Applicant submits data via secure portal
- Backend validates and preprocesses input
- ML engine generates approval probability
- Officer reviews recommendation and decides
- Decision stored; applicant notified instantly

Dataset & Feature Engineering

Core Input Features

The model uses structured applicant data drawn from credit bureaus, application forms, and institutional records.

Preprocessing Pipeline



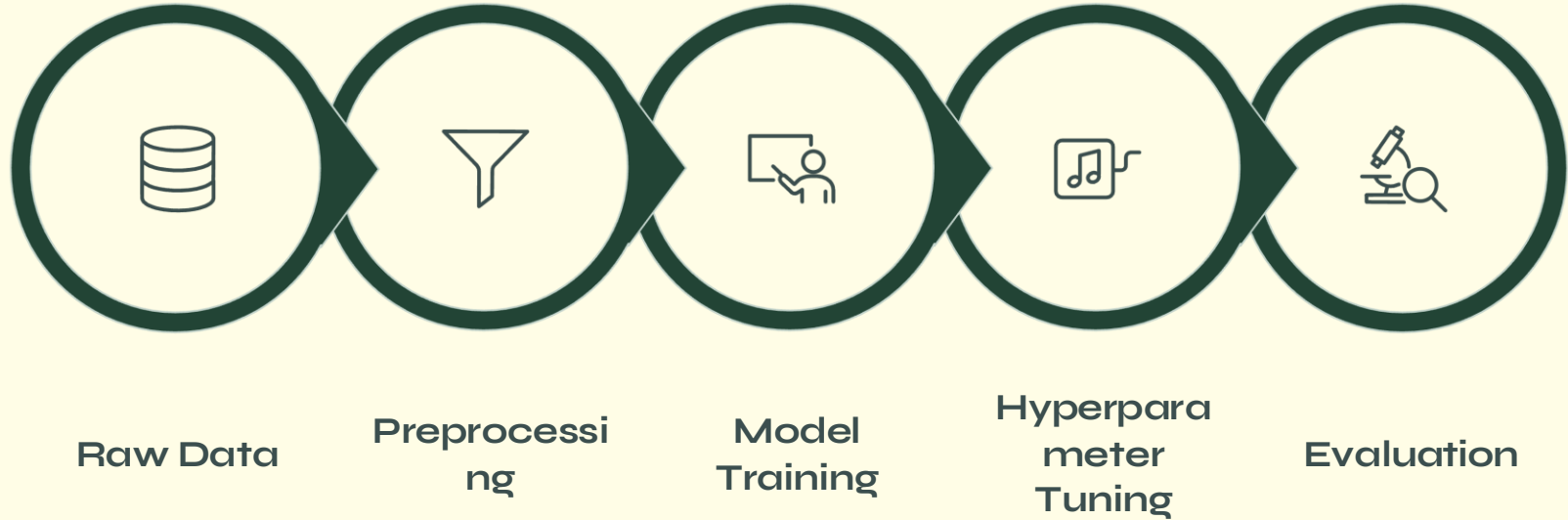
Stratified splitting preserves class balance; correlation matrices and VIF scores eliminate multicollinearity before training.

Financial: Applicant income, co-applicant income, loan amount, loan term

Credit: Credit history (0/1), existing liabilities, repayment capacity

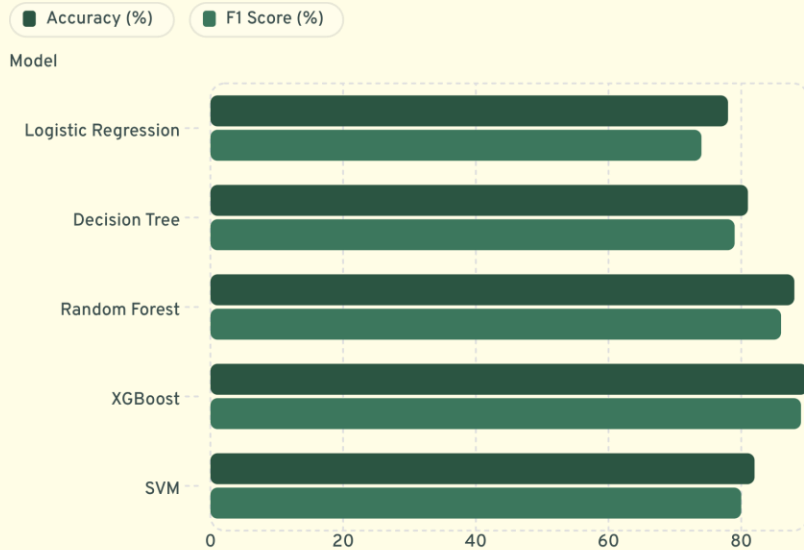
Personal: Education, employment status, marital status, dependents, property area

Machine Learning Pipeline



Five classifiers are trained and compared. **Random Forest** and **XGBoost** consistently outperform linear models on imbalanced lending data, with ensemble methods reducing variance and capturing non-linear feature interactions.

Model Evaluation & Comparison



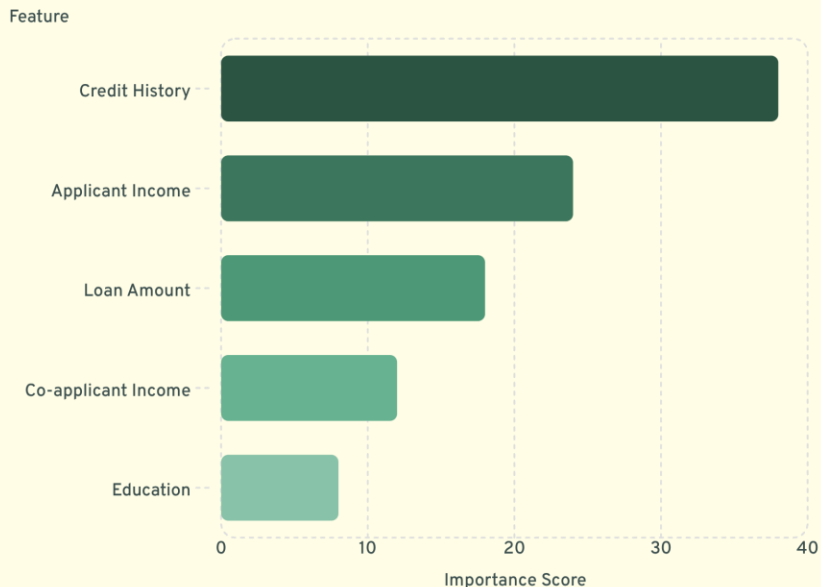
Performance Highlights

XGBoost achieves the highest accuracy (90%) and F1 score (89%), making it the recommended production model. Random Forest is a strong alternative with better interpretability.



Confusion Matrix (XGBoost): TP=142, TN=98, FP=12, FN=8 – indicating low false-negative rate, critical for minimizing credit risk.

Feature Importance Analysis



What Drives the Decision?

Credit history dominates at 38% — a strong binary signal of past repayment behavior. **Applicant income** and **loan amount** jointly determine debt-to-income ratio, the second most critical risk factor.

Lower-ranked features (education, marital status, dependents) provide contextual nuance but rarely override credit and income signals in the final prediction.

User Roles & System Modules



Applicant

Submits applications, uploads documents, tracks real-time status, and receives automated approval or rejection notifications.



Bank Officer

Validates submissions, reviews ML risk scores, overrides decisions when needed, and manages the approval queue.



Administrator

Manages users and permissions, monitors system health, generates compliance reports, and retrains models with new data.



Security, Database & Technology Stack



Authentication

JWT-based sessions with role-based access control (RBAC); bcrypt password hashing; optional 2FA for officers.



Database Design

PostgreSQL with normalized schema: *users*, *applications*, *predictions*, *documents*, *audit_logs*. Encrypted at rest via AES-256.



Tech Stack

Python (scikit-learn, XGBoost, pandas), Flask/FastAPI backend, React frontend, PostgreSQL, Docker for deployment.



Analytics Dashboard

Real-time approval rates, default risk trends, feature drift monitoring, and exportable PDF compliance reports.

Outcomes, Limitations & Future Directions

Expected Advantages

-  **Speed:** Decisions in seconds vs. days
-  **Consistency:** Uniform criteria across all applications
-  **Scalability:** Handles thousands of applications without added staff
-  **Risk Reduction:** Lower default rates through data-driven scoring

Limitations & Roadmap

Current Limits: Model bias risk, cold-start for new applicant profiles, dependency on data quality

Future Enhancements: Deep learning (LSTM for time-series credit behavior), explainable AI (SHAP/LIME), real-time credit bureau API integration, mobile-first UX

Real-World Fit: Deployable in commercial banks, credit unions, fintech lenders, and microfinance institutions